Image Analysis Report

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Executive Summary

The structural base plate assembly exhibits high severity corrosion localized to the anchor bolt fasteners. While the galvanized coating on the main plate and stiffeners remains mostly intact, the degradation of the primary load-bearing fasteners poses a significant risk to structural integrity.

Key Findings

#	Finding
1	Severe atmospheric corrosion and heavy oxide scaling on all three visible double-nut anchor assemblies.
2	Significant 'rust bleeding' and staining extending from the fasteners onto the vertical stiffeners and base plate surface.
3	Apparent material loss on the nut corners, potentially compromising the ability to apply torque or remove them via standard
4	Accumulation of organic debris and grass at the base of the plate, which traps moisture and accelerates localized corrosion
5	The galvanized coating on the base plate is beginning to fail at the interface where it meets the corroded fasteners.
6	Condition of the anchor bolt threads and shank diameter beneath the nuts cannot be determined from the image.
7	Integrity of the grout or concrete interface beneath the base plate is obscured and requires further investigation.

Recommendations

#	Recommendation
1	Perform Ultrasonic Testing (UT) on anchor bolts within 30 days to check for cross-sectional area loss or cracks below the s
2	Mechanically clean fasteners to SSPC-SP3 (Power Tool Cleaning) to remove scale and assess the true remaining metal th
3	Replace all heavily corroded nuts and bolts with new high-strength fasteners that are hot-dip galvanized or coated with a flu
4	Apply a high-build epoxy primer followed by a UV-resistant polyurethane topcoat to the base plate and stiffeners after remo
5	Install plastic or rubber nut caps filled with corrosion-inhibiting grease over new fasteners to prevent future atmospheric exp
6	Clear all vegetation and soil from the perimeter of the base plate to a radius of 1 meter to improve drainage.
7	If section loss on the bolts exceeds 10%, consult a structural engineer for potential supplemental anchor installation or plate
8	Update the asset integrity management system to include biannual visual inspections of this base assembly.